System Design for Modern Microservices

Four quizzes, one at the end of each session. Five questions each.

Tick one answer per question. Mark your own sheet — the answers are read out after each quiz.

Score yourself: 5/5 excellent · 4/5 solid · 3/5 review the notes.

Name: _____ Total: _____ / 20 Date: ____/____/_____

Quiz 1 — Introduction to Microservices

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After Session 1

1. What most accurately defines a monolith?

- ◯ **a)** One codebase deployed as a single unit
- ◯ **b)** Badly written or poorly structured code
- ◯ **c)** An application that does not use a database
- ◯ **d)** Any application older than five years

2. A team of four is building their first product and has no users yet. What is usually the right choice?

- ◯ **a)** Microservices, so they never have to go through a migration later on
- ◯ **b)** Microservices, because that is what modern engineering teams do
- ◯ **c)** A monolith, because it is simpler and they do not yet know the boundaries
- ◯ **d)** One service per developer, so everyone owns their own deployable

3. Two features always change together and always ship together. They should most likely be:

- ◯ **a)** In two services, so they can stay independent
- ◯ **b)** In the same service, deployed as one unit
- ◯ **c)** In two services sharing a single database
- ◯ **d)** In the API gateway, which calls both

4. Which is a genuine advantage of microservices?

- ◯ **a)** There is less total code to write than in a monolith
- ◯ **b)** There are fewer moving parts that can fail at runtime
- ◯ **c)** Debugging a request across the system becomes easier
- ◯ **d)** You can scale and deploy one part without touching the rest

5. “Database per service” means:

- ◯ **a)** Every service is required to use a different database product from all of the others
- ◯ **b)** Each service owns its own data and other services cannot reach into it directly
- ◯ **c)** Each service needs its own physical server to store that data on
- ◯ **d)** Every database must be duplicated so that a backup always exists

Quiz 2 — Architectural Patterns

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After Session 2 · Name: _____

1. What is the main job of an API Gateway?

- ◯ **a)** To store the application's data on behalf of every service
- ◯ **b)** To give clients a single entry point in front of many private services
- ◯ **c)** To remove the need for any service to use a database
- ◯ **d)** To compile and package the application before it is deployed to production

2. In `CATALOG_URL: http://catalog:3001`, what is `catalog`?

- ◯ **a)** A fixed IP address, written into the configuration
- ◯ **b)** A folder on disk that the gateway reads from
- ◯ **c)** A service name, resolved by the platform's DNS
- ◯ **d)** A username that the gateway authenticates with

3. You stopped the recommendations service and the shop kept selling books. This is called:

- ◯ **a)** Graceful degradation
- ◯ **b)** Load balancing
- ◯ **c)** Horizontal scaling
- ◯ **d)** Service discovery

4. Why does the gateway put a 1-second timeout on the recommendations call?

- ◯ **a)** To reduce the amount of network traffic the system pays for
- ◯ **b)** To keep the gateway code shorter and easier to read
- ◯ **c)** So one slow service cannot make every page slow and eventually take the site down
- ◯ **d)** Because the HTTP specification requires that every request carries a timeout value

5. Stopping `catalog` broke the whole page, but stopping `recommendations` did not. The reason is:

- ◯ **a)** Catalog was written in a different programming language from the other three services
- ◯ **b)** Catalog is essential, and the team decided no useful fallback exists for it
- ◯ **c)** Recommendations has fewer lines of code, so it matters less to the system
- ◯ **d)** Catalog was started first, so it holds the network connection open

Quiz 3 — Containerization & Orchestration

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After Session 3 · Name: _____

1. What is the relationship between an image and a container?

- ◯ **a)** They are two words for the same thing
- ◯ **b)** An image is a running copy of a container
- ◯ **c)** A container is a running copy of an image
- ◯ **d)** An image runs inside a container

2. In a Dockerfile, what does `FROM node:22-alpine` do?

- ◯ **a)** Downloads your source code into the image being built
- ◯ **b)** Starts from an existing image that already has Node installed
- ◯ **c)** Sets the port that the application will listen on
- ◯ **d)** Runs the application once the image has finished building successfully

3. You killed a container in this workshop's Compose setup and it stayed dead. Why?

- ◯ **a)** The image became corrupted at the moment the container was killed
- ◯ **b)** Compose needs root privileges before it is allowed to restart anything
- ◯ **c)** Compose only restarts stopped containers on a fixed schedule that it runs periodically
- ◯ **d)** No restart policy is set, and Compose will not maintain a number of healthy copies

4. Which correctly matches the Kubernetes objects?

- ◯ **a)** Pod = a running container · Deployment = keeps N pods alive · Service = a stable address
- ◯ **b)** Pod = a physical machine · Deployment = a container · Service = a database
- ◯ **c)** Pod = a stable address · Deployment = a physical machine · Service = a running container
- ◯ **d)** They are three different names for the same underlying object in the API

5. The gateway's `CATALOG_URL` was unchanged when you moved from Compose to Kubernetes. Why does that matter?

- ◯ **a)** It does not matter, it was a coincidence of how the lab was written
- ◯ **b)** Because both platforms resolve service names, so the application code is portable
- ◯ **c)** Because Kubernetes reads the same docker-compose.yml file directly when it starts up
- ◯ **d)** Because the IP address of the catalog service happened to stay the same

Quiz 4 — Practices & Production

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After Session 4 · Name: _____

1. You deleted a pod and a new one appeared by itself. Which object made that happen?

- ◯ **a)** The Pod
- ◯ **b)** The Service
- ◯ **c)** The Deployment
- ◯ **d)** The container image

2. "Desired state" means:

- ◯ **a)** The state that the system was in the last time it was known to be working correctly
- ◯ **b)** You declare what you want, and the platform continuously repairs reality to match
- ◯ **c)** A backup of the cluster taken immediately before a deployment
- ◯ **d)** The final step of a rolling update, once every pod is replaced

3. You scaled `catalog` to 5 and left `orders` at 1. What is the point?

- ◯ **a)** Catalog is a far more important service than orders, so it deserves more of the cluster resources
- ◯ **b)** Kubernetes schedules more efficiently when a Deployment has an odd number of pods
- ◯ **c)** You can give capacity to the busy part without paying for the rest — impossible in a monolith
- ◯ **d)** Having more pods of any service makes every future deployment complete faster

4. What does a rolling update do?

- ◯ **a)** Replaces instances gradually so the service stays available throughout
- ◯ **b)** Stops every instance, then starts them all again on the new version
- ◯ **c)** Rolls the database back to the state it was in the previous day
- ◯ **d)** Applies updates only during a scheduled weekend maintenance window

5. Which is the strongest reason NOT to adopt microservices?

- ◯ **a)** Writing the application code itself takes noticeably longer than it would in an equivalent monolith
- ◯ **b)** They use more disk space and memory than an equivalent monolith would use
- ◯ **c)** Your team is small, the boundaries are unclear, and the operational cost exceeds the benefit
- ◯ **d)** They cannot be run at all without Kubernetes or a managed cloud provider

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